

Neel M Naik

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PROFILE OVERVIEW

Results-driven Information Science Engineering student specializing in machine learning, 3D computer vision, and algorithmic optimization. Proven expertise in developing PyTorch models for LiDAR point-cloud processing, alongside engineering high-contention scheduling engines and secure backend utilities. Adept at transforming complex spatial datasets and strict constraints into scalable, interactive software solutions.

EDUCATION

Kendriya Vidyalaya, Bengaluru Class X, CBSE, 5.8 CGPA	2007 – 2017
Narayana PU College, Bengaluru Class XII, PCMB, 43.67%	2018 – 2019
Nitte Meenakshi Institute of Technology, Bengaluru B.E. in Information Science and Engineering — [CGPA / Percentage]	2021 – Present

WORK EXPERIENCE

Intern — Bharat Electronics Limited (BEL), Bengaluru Product Development and Innovation Centre (PDIC) - Worked on a project titled “Secure File Encryption Utility” — addressed the critical risk of exposed local data by developing a Python encryption tool that leverages the Advanced Encryption Standard to scramble plaintext into unreadable ciphertext. - Developed a standalone file security tool using Python and the cryptography package to execute AES-128 symmetric encryption, secure key storage, and lossless decryption for localized plaintext files. - Successfully validated lossless data restoration while mastering symmetric key lifecycles, the CIA triad, and defense-standard data handling protocols.	Oct 2025 – Nov 2025
Machine Learning Engineer, Intern — Bellatrix Aerospace, Bengaluru - Engineered a 3D point-cloud registration pipeline using PointNet autoencoders, PCA, and ICP algorithms to identify and estimate satellite poses from LiDAR spatial datasets. - Processed and trained models on datasets containing over 20,000+ spatial data points, reducing rotational pose estimation errors to < 4 degrees to support autonomous rendezvous simulations.	Mar 2026 – Jun 2026

PROJECTS CREATED

Interactive 3D Virtual Environment C#, Unity Engine, 3D Spatial Design - Developed an immersive, interactive 3D virtual environment from scratch using the Unity engine and object-oriented C# scripting. - Programmed custom user controls, player kinematics, and real-time environmental physics to create a seamless and responsive user experience. - Designed and optimized spatial layouts and rendering pipelines to maintain high frame rates and smooth interactions within the simulated world.	Jul 2024
Secure File Encryption Utility Python 3.x, Visual Studio Code, Git - Engineered a standalone file encryption utility in Python using the cryptography library (Fernet/AES-128) to secure local plaintext data at rest against unauthorized physical access. - Implemented automated 128-bit symmetric key generation, safe key storage, and binary file I/O operations for seamless file overwriting. - Validated end-to-end encryption and decryption pipelines, ensuring 100% data confidentiality and zero data corruption during data restoration.	Nov 2025
Satellite Identification via 3D Point Clouds Python, PyTorch, PointNet, PCA, ICP, Google Colab - Engineered an end-to-end machine learning pipeline utilizing PointNet autoencoders to accurately classify and register 3D satellite models from raw LiDAR spatial datasets. - Implemented Principal Component Analysis (PCA) and Iterative Closest Point (ICP) algorithms from scratch to optimize point-cloud alignment and 3D reconstruction accuracy. - Processed high-dimensional spatial data within Google Colab, successfully reducing point-cloud registration errors and ensuring high-fidelity rendering for aerospace simulations.	Jun 2026
Dynamic Placement Week Scheduler & Replanning Engine Python 3.11, Streamlit, Pandas, Heuristic Optimization, Constraint Satisfaction Problems (CSP), OOP - Engineered a discrete-event heuristic scheduling engine coordinating high-contention campus recruitment drives across 35 companies, 800 candidates, and 20 interview rooms over a 4-day period. - Implemented priority-driven slot allocation algorithms enforcing strict hard constraints, achieving zero double-booking across candidates, interview rooms, and multi-member evaluation panels. - Developed a real-time replanning mechanism to dynamically handle operational disruptions (delayed recruiters, panel dropouts, room outages, and candidate withdrawals) while bounding schedule reshuffling strictly to affected entities. - Built and deployed an interactive Streamlit control room featuring automated constraint-verification test suites, live conflict detection, and structured diff logging, ensuring 100% mathematical clash prevention.	Aug 2026

SKILLS

Technical Skills: Python, PyTorch, C#, C, SQL, Deep Learning (PointNet, CNNs, Autoencoders), 3D Computer Vision (ICP, PCA, LiDAR Processing), Cryptography (AES-256), Heuristic Optimization, Constraint Satisfaction Problems (CSP), Object-Oriented Programming (OOP), Data Structures & Algorithms
Tools & Platforms: Git, GitHub, Google Colab, VS Code, Unity Engine, Streamlit, Linux/Unix, LaTeX, AWS (Cloud Fundamentals)
Soft Skills: Complex Problem-Solving, Analytical Thinking, Cross-Functional Collaboration, Technical Documentation & Reporting, Agile Adaptability, System Design & Planning

ACTIVITIES

Strength Training Dedicated to a multi-year fitness and strength training regimen.	2018 – Present
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